

Four Sizes of the Rational Numbers

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1 The question is not one question

The rational numbers are dense in the real line, and so are the irrationals. The rationals are countable, and the irrationals are not. These facts are usually assembled into a slogan: there are *infinitely more* irrationals than rationals. The slogan is not merely loose. It is not a statement, because “infinitely more” names no operation on cardinals, and, as will appear in Section 4, the question it gestures at has no answer inside ZFC.

The purpose of this essay is to replace the slogan with something better. The claim to be defended is that *size* of a subset of $\mathbb{R}$ is not a single notion but at least four inequivalent ones — density, cardinality, measure, and category — and that $\mathbb{Q}$ is the example on which they visibly come apart. Under three of them $\mathbb{Q}$ is as small as a nonempty set can be. Under the fourth it is as large as any subset of $\mathbb{R}$ can be. No single intuition survives this.

2 Setting the two density claims straight

The density statements are frequently quoted in a weakened form: between any two rationals there is an irrational, and between any two irrationals there is a rational. Both are true, but neither is the property being named. Density of a set $S \subseteq \mathbb{R}$ means that every nonempty open interval of $\mathbb{R}$ meets S — the endpoints range over all of $\mathbb{R}$, not over a distinguished subset. The restricted versions happen to be equivalent to the real ones, but only by a detour through the real ones, which makes them a poor way to state the theorem.

Theorem 2.1 (Density of $\mathbb{Q}$). *For all real $a < b$ there is $q \in \mathbb{Q}$ with $a < q < b$.*

Proof. By the Archimedean property choose $n \in \mathbb{N}$ with $n(b - a) > 1$, so that $1/n < b - a$. Let $m = \lfloor na \rfloor + 1$, so that $m - 1 \leq na < m$. Then $a < m/n$, and

$$\frac{m}{n} \leq \frac{na + 1}{n} = a + \frac{1}{n} < a + (b - a) = b.$$

Hence $q = m/n$ works. □

Theorem 2.2 (Density of $\mathbb{R} \setminus \mathbb{Q}$). *For all real $a < b$ there is $x \in \mathbb{R} \setminus \mathbb{Q}$ with $a < x < b$.*

Proof. Apply the previous theorem to the interval $(a - \sqrt{2}, b - \sqrt{2})$, obtaining $q \in \mathbb{Q}$ with $a - \sqrt{2} < q < b - \sqrt{2}$. Then $q + \sqrt{2} \in (a, b)$, and it is irrational: were $q + \sqrt{2} = r \in \mathbb{Q}$, we would have $\sqrt{2} = r - q \in \mathbb{Q}$. □

The second proof is worth a moment. It uses nothing about $\mathbb{Q}$ except that it is a dense subgroup of $(\mathbb{R}, +)$ and that its complement is nonempty; the translate of a dense set by a fixed real is dense, and the translation carries $\mathbb{Q}$ off itself. The same argument shows that any dense set has dense complement whenever the complement contains a coset of the set. This is the first sign that density is a coarse instrument.

3 Density carries no size information at all

Proposition 3.1. *Density is independent of cardinality: each of the four combinations of (countable, uncountable) with (dense, nowhere dense) is realised by a subset of $\mathbb{R}$.*

Discussion. $\mathbb{Q}$ is countable and dense. $\mathbb{R} \setminus \mathbb{Q}$ is uncountable and dense. The set $\mathbb{Z}$ is countable and nowhere dense. The Cantor middle-thirds set C is uncountable — it surjects onto $\{0, 1\}^{\mathbb{N}}$ by ternary expansion — and nowhere dense, being closed with empty interior. $\square$

So the observation that $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are both dense tells us precisely nothing about their relative size, and one should resist the common instinct to treat “dense” as a largeness property at all. Density is a statement about closure: S is dense exactly when $\overline{S} = \mathbb{R}$. It says that S approximates every real arbitrarily well. It does not say that S has many elements, and the Cantor set shows the converse failure — a set can have the cardinality of the continuum while approximating almost nothing.

4 Cardinality is an equivalence class, not a magnitude

Theorem 4.1. *$\mathbb{Q}$ is countable and $\mathbb{R} \setminus \mathbb{Q}$ is not.*

Proof. Each nonzero $q \in \mathbb{Q}$ has a unique representation $\pm m/n$ in lowest terms with $m, n \in \mathbb{N}$, which gives an injection $\mathbb{Q} \rightarrow \mathbb{Z} \times \mathbb{N}$; the latter is countable, hence so is $\mathbb{Q}$. For the second claim, suppose $\mathbb{R} \setminus \mathbb{Q}$ were countable. Since $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ is then a union of two countable sets, $\mathbb{R}$ would be countable, contradicting Cantor’s theorem that no surjection $\mathbb{N} \rightarrow \mathbb{R}$ exists. $\square$

Corollary 4.2. $|\mathbb{R} \setminus \mathbb{Q}| = |\mathbb{R}| = 2^{\aleph_0}$.

Now the point of the section. The relation $\aleph_0 < 2^{\aleph_0}$ unpacks to exactly two assertions: there is an injection $\mathbb{Q} \rightarrow \mathbb{R} \setminus \mathbb{Q}$, and there is no bijection. Nothing further is being claimed, and in particular no quantity has been produced that measures the gap. The temptation to ask “how much bigger” has two independent obstructions.

The first is algebraic. Cardinal addition absorbs: $\aleph_0 + 2^{\aleph_0} = 2^{\aleph_0}$, which is just the statement that adjoining $\mathbb{Q}$ back to $\mathbb{R} \setminus \mathbb{Q}$ recovers $\mathbb{R}$ without changing its cardinality. Because addition absorbs, it is not cancellative, and there is no subtraction. The expression “how many more irrationals than rationals” is therefore not an expression.

The second obstruction is deeper. Even the question of *where* $2^{\aleph_0}$ sits in the hierarchy of alephs is not settled by the axioms. The continuum hypothesis, $2^{\aleph_0} = \aleph_1$, was shown consistent with ZFC by Gödel (1940) and its negation shown consistent by Cohen (1963). So “how much larger is the continuum than $\aleph_0$ ” has no answer in ZFC — not an unknown answer, but no answer determined by the theory. A phrase like “infinitely more” does not merely overstate what has been proved; it presupposes a determinate quantity that the axioms decline to fix.

What cardinality does give is a genuine and useful dichotomy: countable sets can be enumerated, indexed, and exhausted by induction; uncountable ones cannot. That is a structural fact with real consequences, and it is all that $\aleph_0 < 2^{\aleph_0}$ asserts.

5 Measure supplies what cardinality withholds

Cardinality is invariant under arbitrary bijections and so cannot see the geometry of the line. Lebesgue measure can.

Theorem 5.1. $\mathbb{Q}$ has Lebesgue measure zero.

Proof. Enumerate $\mathbb{Q} = \{q_1, q_2, \dots\}$ and fix $\varepsilon > 0$. Put $I_k = (q_k - \varepsilon 2^{-k-1}, q_k + \varepsilon 2^{-k-1})$. Then $\mathbb{Q} \subseteq \bigcup_k I_k$ and

$$\sum_{k=1}^{\infty} |I_k| = \sum_{k=1}^{\infty} \varepsilon 2^{-k} = \varepsilon.$$

Since ε was arbitrary, the outer measure of $\mathbb{Q}$ is 0. □

Corollary 5.2. $\lambda((\mathbb{R} \setminus \mathbb{Q}) \cap [0, 1]) = 1$.

This is the theorem behind the slogan “almost every real is irrational,” and unlike the slogan about cardinality it has exact content: a point chosen uniformly at random from $[0, 1]$ lies in $\mathbb{Q}$ with probability zero. Note also what the covering argument used — only that $\mathbb{Q}$ is countable. Every countable set is null. The implication is one-directional, and this matters:

Example 5.3. The Cantor set C is null. Its complement in $[0, 1]$ has measure $\sum_{n \geq 0} 2^n 3^{-n-1} = 1$. Yet $|C| = 2^{\aleph_0}$.

So measure and cardinality separate in both directions: $\mathbb{Q}$ is small in both senses, C is null but of full cardinality, and $\mathbb{R} \setminus \mathbb{Q}$ is large in both. Three of the four combinations already occur among sets we have named. A “size” that assigns $\mathbb{Q}$ and C the same value while cardinality distinguishes them, and distinguishes C from $[0, 1]$ while cardinality does not, is plainly measuring something else.

6 Category gives a third answer, and it is incompatible

Definition 6.1. A set is *meagre* (first category) if it is a countable union of nowhere dense sets, and *comeagre* if its complement is meagre.

Proposition 6.2. $\mathbb{Q}$ is meagre and $\mathbb{R} \setminus \mathbb{Q}$ is comeagre.

Proof. $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$, a countable union of singletons, each nowhere dense. □

By the Baire category theorem $\mathbb{R}$ is not meagre, so the notion is not vacuous: meagre is a genuine smallness property, and it again places $\mathbb{Q}$ on the small side. Three notions, three verdicts, all agreeing that $\mathbb{Q}$ is negligible. It would be natural to conclude that measure and category are two routes to the same underlying idea. They are not.

Theorem 6.3 (Erdős–Sierpiński duality, decomposition form). *There is a partition $\mathbb{R} = A \sqcup B$ with A meagre and $\lambda(B) = 0$.*

Sketch. Enumerate $\mathbb{Q} = \{q_k\}$ and for each n let $U_n = \bigcup_k (q_k - 2^{-k-n}, q_k + 2^{-k-n})$, an open dense set of measure at most 2^{1-n} . Put $B = \bigcap_n U_n$. Then $\lambda(B) = 0$, while $A = \mathbb{R} \setminus B = \bigcup_n (\mathbb{R} \setminus U_n)$ is a countable union of closed sets with empty interior, hence meagre. □

The consequence is worth stating flatly. Every real number lies either in a set that topology calls negligible or in a set that measure calls negligible. The two notions of smallness are not refinements of one another; they are, on the line, complementary. Any argument of the form “this set is small, therefore its complement is large” must therefore name which smallness it means, and cannot silently transfer the conclusion to the other.

7 A fourth notion, for contrast

Hausdorff dimension refines measure by asking how the covering cost scales. For $s \geq 0$ one sets

$$\mathcal{H}^s(E) = \lim_{\delta \rightarrow 0} \inf \left\{ \sum_i (\text{diam } U_i)^s : E \subseteq \bigcup_i U_i, \text{diam } U_i < \delta \right\},$$

and $\dim_H E$ is the critical exponent at which $\mathcal{H}^s(E)$ drops from ∞ to 0. Every countable set has dimension 0, so $\dim_H \mathbb{Q} = 0$. But $\dim_H C = \log 2 / \log 3 \approx 0.631$. Lebesgue measure assigns $\mathbb{Q}$ and C the same value; dimension separates them. Where cardinality was too coarse to see geometry, measure is too coarse to see scaling.

8 What each notion is for

The four notions are not competitors, and none is the “real” one. They answer different questions, and the right one is fixed by the use.

- *Density* asks whether the set approximates every point. It is the right notion when the object of interest is a continuous function, since a continuous function is determined by its values on a dense set — which is exactly why $\mathbb{Q}$ being countable and dense makes $\mathbb{R}$ separable and makes numerical analysis possible.
- *Cardinality* asks whether the set can be enumerated. It is the right notion when the argument proceeds by listing, indexing, or induction along an enumeration — as in the covering proof above, which needed countability and nothing else.
- *Measure* asks what the probability of landing in the set is. It is the right notion for anything probabilistic or integral-theoretic, and it is what licenses “almost every.”
- *Category* asks whether the set is topologically typical. It is the right notion for genericity arguments — the standard proof that the generic continuous function on $[0, 1]$ is nowhere differentiable is a category argument, and no measure argument substitutes for it.

9 Conclusion

The four opening claims are true, but the moral usually drawn from them is not. It is not that the irrationals outnumber the rationals by some infinite factor; that statement has no referent, and the quantity it implies is independent of the axioms. The moral is that $\mathbb{Q}$ is a set on which our notions of size visibly disagree: dense, hence topologically ubiquitous; countable, hence enumerable; null, hence probabilistically invisible; meagre, hence topologically negligible — and, by the duality theorem, negligibility in the last two senses is not one property but two opposed ones.

The instinct being corrected is the instinct to hear “smaller” as a single word. On subsets of the real line it is at least four words, and the rationals are where the difference is impossible to overlook.